

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1704

COURSE OUTCOME COVERED

***C04:** Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I MITHRA DEVI S - 192524203 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Mithra Devi

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Task 1: Article Summary

Citation Judy Simon, Nellore Kapileswar, and Anoop Mohanakumar, “A Novel Energy Adaptive Neural Network and Deep Q-Learning Network for Improved Energy Efficiency in Dynamic Underwater IoT Environment,” IEEE Access, vol. 13, pp. 141403–141418, Aug. 2025. DOI: 10.1109/ACCESS.2025.3597025 ([doi.org in Bing](https://doi.org/10.1109/ACCESS.2025.3597025))

Domain/Application Area Underwater Internet of Things (UIoT) — energy-efficient and reliable communication in hybrid optical-acoustic networks.

Summary (≈180 words) The article addresses the challenge of energy efficiency and communication reliability in dynamic underwater IoT environments. Traditional clustering and routing methods fail to adapt to fluctuating underwater conditions, leading to high energy consumption and unreliable data transmission. The authors identify a research gap: most prior works treat power control and modulation separately, lacking a unified adaptive framework. To bridge this, they propose two integrated models — an Energy Adaptive Neural Network (EANN) for dynamic transmission power and frequency selection, and a Deep Q-Learning Adaptive Error Correction and Modulation (DQL-AECM) system for real-time modulation and error correction. Together, these models form a joint deep learning framework that simultaneously optimizes energy usage and communication reliability. Experimental results show significant improvements over classical protocols (LEACH, PEGASIS, TEEN, MTP), achieving lower energy consumption (22 J), reduced BER (10^{-8}), higher throughput (6.2 Mbps), and lower latency (35 ms). This work contributes a scalable, adaptive solution for underwater IoT systems, ensuring robust performance in highly variable marine environments.

Task 2: Methodology Analysis

- Techniques Used
 - EANN: Neural network for adaptive power control and frequency selection.
 - DQL-AECM: Deep Q-Learning reinforcement model for adaptive modulation and error correction.
- Dataset
 - Simulated underwater IoT network in Aqua-Sim.
 - Nodes modeled with hybrid optical-acoustic communication.
 - Channel state information (attenuation, SNR, BER, noise power) used as input features.
 - No real-world dataset; simulation-based evaluation.
- Mapping to CO4 Topics

- Reinforcement Learning → DQL-AECM.
- Inductive Learning → EANN (neural network adaptation).
- Strength
 - Joint integration of power control and modulation adaptation — addresses interdependence often ignored in prior works.
- Limitation
 - Simulation-based evaluation only; lacks validation on real underwater deployments.

Task 3: Results & Critical Evaluation

Key Results

<u>Model</u>	<u>Energy (J)</u>	<u>BER</u>	<u>SNR (dB)</u>	<u>Throughput (Mbps)</u>	<u>Latency (ms)</u>
<u>Proposed EANN+DQL</u>	<u>22</u>	<u>1×10^{-8}</u>	<u>21</u>	<u>6.2</u>	<u>35</u>
<u>LEACH</u>	<u>35</u>	<u>1×10^{-6}</u>	<u>15</u>	<u>4.1</u>	<u>60</u>
<u>PEGASIS</u>	<u>32</u>	<u>1×10^{-6}</u>	<u>16</u>	<u>4.5</u>	<u>55</u>
<u>TEEN</u>	<u>30</u>	<u>1×10^{-7}</u>	<u>17</u>	<u>4.8</u>	<u>50</u>
<u>MTP</u>	<u>28</u>	<u>1×10^{-7}</u>	<u>18</u>	<u>5.0</u>	<u>48</u>

Critical Evaluation

- Metrics are promising but rely on controlled simulations; real-world underwater conditions (turbidity, currents) may degrade performance.
- Potential bias: dataset limited to Aqua-Sim scenarios, not diverse oceanic environments.
- Additional experiments: field trials with real sensor nodes, testing scalability in large networks, and robustness under extreme noise conditions.

Real-World Applicability

- Applications: marine life monitoring, pollution detection, underwater exploration.
- Challenges:
 - Data availability: real underwater datasets are scarce.
 - Compute: reinforcement learning requires significant resources.

- Ethics: deployment in sensitive ecosystems must avoid ecological disruption.

Task 4: Reflection & Learning Outcome

Three Key Insights

1. Adaptive Power Control (EANN) → Deepened understanding of *Inductive Learning* applied to dynamic environments.
2. Reinforcement Learning for Modulation (DQL-AECM) → Reinforced concept of *policy optimization* in RL.
3. Joint Optimization Framework → Showed importance of *integrating multiple ML techniques* rather than treating them in isolation.

Extension Proposal

- Novel Experiment: Apply the framework to satellite-assisted ocean monitoring with real-world sensor data.
- Justification: Expands applicability beyond simulations, tests scalability in global networks.
- Expected Impact: More robust, industry-ready solution for large-scale marine monitoring.